Working with OpenSSL

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Overview

OpenSSL Certificates

- Private key
- Certificate Signing Request
- Certificate File
- SAN - or Subject Alt Name
  RFC 6125
- Certificate Chains
- Nginx and HTTPS

OpenSSL

```
$ openssl genpkey -algorithm RSA -out private.key
```

OpenSSL Private Key

To begin, we must generate a private key

```
$ openssl req -new -key private.key -out csr.pem
```

OpenSSL Certificate Signing Request

We can then generate the public element of the private key by creating a certificate signing request to be forwarded to a trusted CA or Certificate Authority (such as Let's Encrypt). This will prompt for details to add to the certificate, including the CN or Common Name. The CN is deprecated in favor of the Subject Alternate Name which is not prompted for. Better we use a file to provide answers....

openssl.conf

```
[req]
prompt = no
distinguished_name = req_distinguished_name
req_extensions = v3_req

[req_distinguished_name]
countryName = GB
stateOrProvinceName = London
localityName = Westminster
organizationName = Example Widgets
organizationalUnitName = IT
commonName = www.example.com
emailAddress = admin@example.com

[v3_req]
subjectAltName = @alt_names

[alt_names]
DNS.1 = example.com
DNS.2 = www.example.com
```

```
$ openssl req -new -key private.key -out csr.pem -config openssl.conf
```

OpenSSL Certificate Signing Request Automated

Having created a configuration file, you can reference it from the command line

```
$ nano v3.ext
keyUsage               = digitalSignature, nonRepudiation, keyEncipherment,
dataEncipherment, keyAgreement, keyCertSign
subjectAltName         = DNS:example.com, DNS:www.example.com
$ openssl x509 -req -in csr.pem  -signkey private.key -out certificate.crt -days 3650 -
sha256 -extfile v3.ext
```

CA Signing

Yoy can sign your own certificate request, a self-signed certificate. To allow for the SAN or Subject AltName, you must reference a configuration.

Demo

Creating a Self-Signed Certifcate

- Tools and configuration files
- RFC 6125 SANs

Reading Certificates

```
$ openssl x509 -in certificate.crt -noout -text
$ openssl s_client -connect pluralsight.com:443 [-showcerts|-debug]
```

Tools to Read Certificate Details

Using OpenSSL we can read local files or certificates from remote servers. Typically, public servers will present a certificate chain showing Cas who have been used to sign server requests. Using CAs the browser or SSL client only needs to trust the CA

Demo

Reading Certificate Details

- openssl x509
- openssl s_client

Nginx Web Server

```
$ sudo apt install nginx w3m
```

Nginx Web Server

You can install the web server and command line web browser

/etc/nginx/sites-available/my-site

```
server {
    listen 80;
    listen [::]:80;
    server_name www.example.com;
    return 301 https://$host$request_uri;
}

server {
    listen 443 ssl;
    listen [::]:443 ssl;
    server_name www.example.com;
    ssl_certificate /ssl/certificate.crt;
    ssl_certificate_key /ssl/private.key;
    location / {
        root /var/www/html;
        index index.html;
    }
}
```

```
$ sudo mkdir /ssl
$ sudo cp ~vagrant/{private.key,certificate.crt} /ssl
$ echo "Hello" | sudo tee /var/www/html/index.html
$ sudo sed -i '$a 127.0.0.1 www.example.com' /etc/hosts
$ sudo rm /etc/nginx/sites-enabled/default
$ sudo ln -s /etc/nginx/sites-available/my-site /etc/nginx/sites-enabled/
$ sudo systemctl restart nginx
$ w3m localhost
```

Final Steps

You can copy the key and certfificate to the correct location refenced by the file, create a welcome page and add the DNS name to to local hosts file. Next you can unlink the original configuration and link the new configuration before restarting nginx. Connecting to HTTP will redirect to HTTPS. You will be warned about the Self-Signed cert and that it is untrusted.

Demo

Testing HTTPS:

- Install Nginx
- Create configuration
- Test HTTPS

Summary

Working with OpenSSL

- OpenSSL
  - genpkey
  - req
  - x509
  - s_client
- Nginx
  - Redirect HTTP
  - Test HTTPS

Up Next:

Congratulations